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Desigualdad de Minkowski

Proposición 1 (Desigualdad de Minkowski). *Sea* $1 \leq p \leq \infty$ *y* $f, g \in \mathcal{L}^p(\mu)$

$$\Longrightarrow \|f+g\|_p \leq \|f\|_p + \|g\|_p .$$

***Demostración*:** Dividimos la demostración en tres casos excluyentes:

I. Si $p = 1$, entonces podemos aplicar directamente la desigualdad triangular de $|\cdot|$:

$$\begin{aligned}\|f+g\|_1 &= \int_\Omega |f(x)+g(x)| \,\mathrm{d}\mu(x) \overset{\overset{\triangle}{\downarrow}}{\leq} \int_\Omega \big[\, |f| + |g| \,\big] \,\mathrm{d}\mu(x) \\ &\overset{\overset{\text{lineal}}{\downarrow}}{=} \int_\Omega |f| \,\mathrm{d}\mu(x) + \int_\Omega |g| \,\mathrm{d}\mu(x) = \|f\|_1 + \|g\|_1 .\end{aligned}$$

II. Si $1 < p < \infty$, entonces

$$\begin{aligned}|f(x)+g(x)|^p &= |f(x)+g(x)| \, |f(x)+g(x)|^{p-1} \\ &= |f(x)| \, |f(x)+g(x)|^{p-1} + |g(x)| \, |f(x)+g(x)|^{p-1} .\end{aligned} \tag{1}$$

Integrando el primer sumando del lado derecho, obtenemos

$$\begin{aligned}\int_\Omega |f(x)| \, |f(x)+g(x)|^{p-1} \,\mathrm{d}\mu(x) &\overset{\overset{\text{Hölder}}{\downarrow}}{\leq} \|f\|_p \left\| |f+g|^{p-1} \right\|_q = \\ &\leq \|f\|_p \left[\int_\Omega |f+g|^{(p-1)q} \,\mathrm{d}P \right]^{1/q} = \\ &\leq \|f\|_p \left[\int_\Omega |f+g|^p \,\mathrm{d}P \right]^{\frac{1}{p} \cdot \frac{p}{q}} = \|f\|_p \Big(\|f+g\|_p \Big)^{p/q} .\end{aligned} \tag{2}$$

Entonces, como consecuencia de (1) y (2), deducimos que

$$\begin{aligned}\|f+g\|_p^p &\overset{\overset{(1)}{\downarrow}}{=} \int_\Omega |f(x)| \, |f(x)+g(x)|^{p-1} \,\mathrm{d}\mu(x) + \int_\Omega |g(x)| \, |f(x)+g(x)|^{p-1} \,\mathrm{d}\mu(x) \\ &\overset{\overset{(2)}{\downarrow}}{\leq} \|f\|_p \|f+g\|_p^{p/q} + \|g\|_p \|f+g\|_p^{p/q} = \|f+g\|_p^{p/q} \Big[\|f\|_p + \|g\|_p \Big] . \\ \Longleftrightarrow \|f+g\|_p^{p-p/q} &= \|f+g\|_p \leq \|f\|_p + \|g\|_p .\end{aligned}$$

III. Si $p = \infty$, entonces tomamos

$$A_1 \subset \Omega : \|f\|_\infty = \sup_{x \in A_1^c} |f(x)| \quad \text{y} \quad A_2 \subset \Omega : \|g\|_\infty = \sup_{x \in A_2^c} |g(x)| \,.$$

Definimos $A := A_1 \cup A_2$, entonces

$$\begin{aligned} \|f\|_\infty + \|g\|_\infty &= \sup_{x \in A_1^c} |f(x)| + \sup_{x \in A_2^c} |g(x)| \geq \sup_{x \in A^c} |f(x)| + \sup_{x \in A^c} |g(x)| \\ &\geq \sup_{x \in A^c} \big[\, |f(x)| + |g(x)| \,\big] \underset{\substack{\uparrow \\ \triangle}}{\geq} \sup_{x \in A^c} |f(x) + g(x)| \\ &\geq \inf_{\substack{B \in \Sigma \\ \mu(B)=0}} \sup_{x \in B^c} |f(x) + g(x)| = \|f + g\|_\infty \,. \end{aligned}$$

Como se ha demostrado en los tres casos, se tiene que $\|f + g\|_p \leq \|f\|_p + \|g\|_p$. ∎

Referenciado en

- `Teo-esp-lp-banach`
- `Ley-debil-grandes-numeros`
- `Lem-esp-lp-vectorial`
- `Lem-esp-lp-normado`